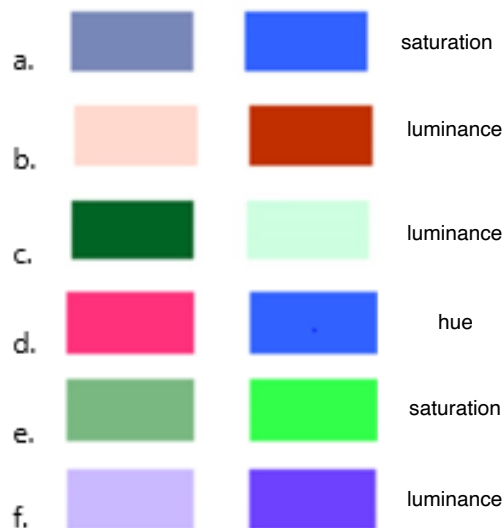


Workshop Activities

Activity 1 – Differentiating Between Hue, Saturation, and Luminance

Each of the following pairs of colours differ on a single characteristic of colour (hue, saturation, or luminance). Each pair of shapes provided below, in parts a through f.



Try to determine whether each pair of colours differs on hue, saturation, or luminance and explain your answer.

Activity 2 – Creating a Choropleth Map using Excel

This activity creates a choropleth map using data from the *StatePopulation* data file. The file *StatePopulation* contains population data for the fifty states of the United States. A sample of the data is shown below.

	A	B
1	State	Estimated 2020 Population
2	California	39,937,489
3	Texas	29,472,295
4	Florida	21,992,985
5	New York	19,440,469
6	Pennsylvania	12,820,878
7	Illinois	12,659,682
8	Ohio	11,747,694
9	Georgia	10,736,059
10	North Carolina	10,611,862

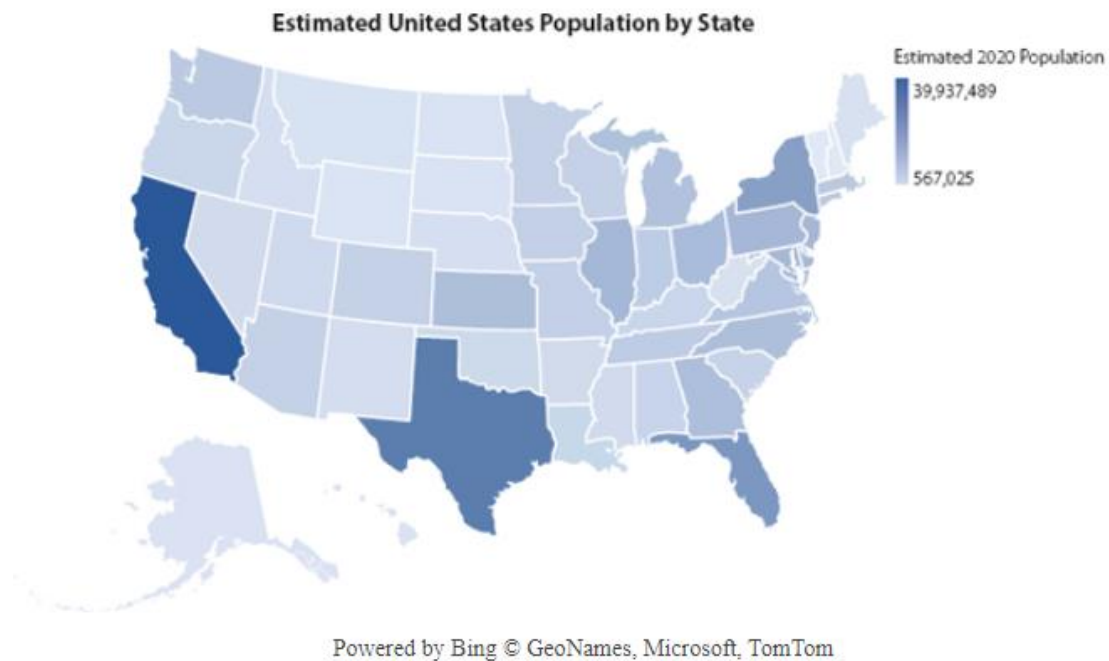
The choropleth map will use colour shading to denote the population of each state. A darker shade will indicate a higher population and a lighter shade will indicate a lower population.

Open the *StatePopulation* data file and complete the following steps:

- Step 1.
 - Select cells A1:B51
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - In the Charts group, click the Maps button
 - Select Filled Map

Then edit the chart to improve its appearance.

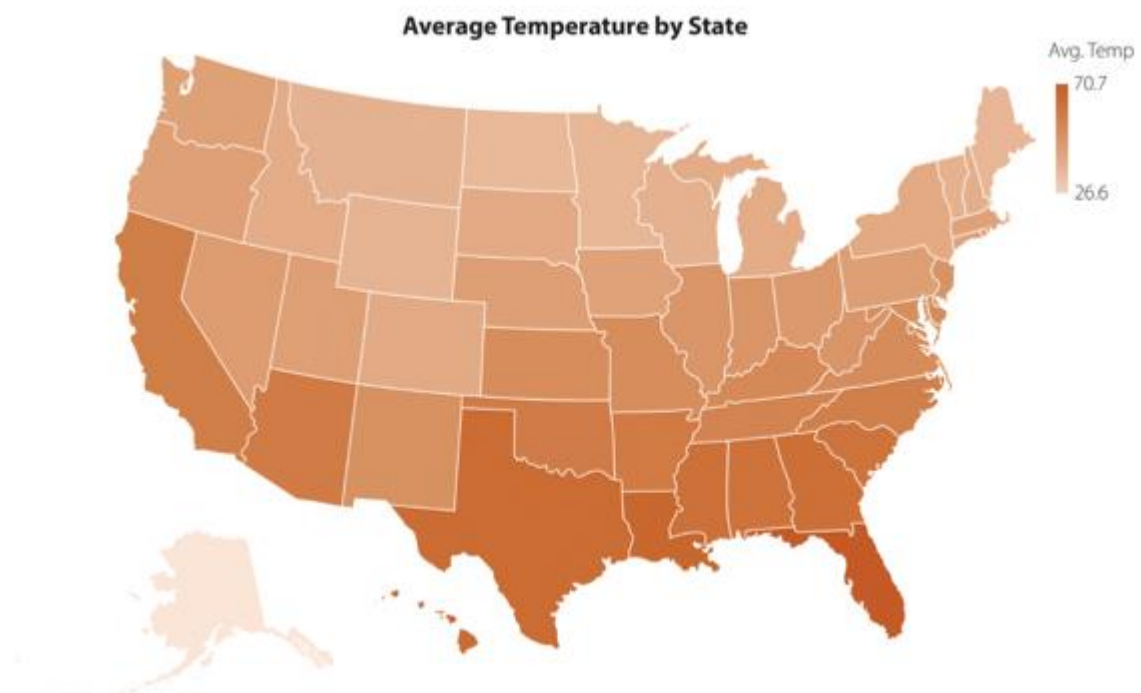
Your finished choropleth map chart should look similar to the following chart.



What does the choropleth map tell us?

Activity 3 – Choosing Colour Schemes

Consider the choropleth maps of average annual temperature by state shown below.



Different levels of luminance for the same hue were used to indicate the gradient of a quantitative variable, temperature.

Suppose you need to create a choropleth map of median family income by state. What colour scheme would you choose to convey the message to the audience and why? Define the colour scheme, hue and luminance selected for this case.

What if instead you need to create a choropleth map of changes in median family income by state over the past decade (such as 2010 vs. 2020.) What colour scheme would you choose to convey that message to the audience and why? Define colour scheme, hues, and luminance selected for this case

Activity 4 – Creating a Heatmap using Excel

This activity creates a heatmap using data from the *SameStoreSales* data file. The rows of the *SameStoreSales* data set correspond to store locations and the columns are the months of the year. The percentages given indicate the change in sales over the same month last year for a given store. This percentage change metric is commonly used in the retail industry and is referred to as “same-store-sales”. The data is shown below.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
2	St. Louis	-2%	-1%	-1%	0%	2%	4%	3%	5%	6%	7%	8%	8%
3	Phoenix	5%	4%	4%	2%	2%	-2%	-5%	-8%	-6%	-5%	-7%	-8%
4	Albany	-5%	-6%	-4%	-5%	-2%	-5%	-5%	-3%	-1%	-2%	-1%	-2%
5	Austin	16%	15%	15%	16%	18%	17%	14%	15%	16%	19%	18%	16%
6	Cincinnati	-9%	-6%	-7%	-3%	3%	6%	8%	11%	10%	11%	13%	11%
7	San Francisco	2%	4%	5%	8%	4%	2%	4%	3%	1%	-1%	1%	2%
8	Seattle	7%	7%	8%	7%	5%	4%	2%	0%	-2%	-4%	-6%	-5%
9	Chicago	5%	3%	2%	6%	8%	7%	8%	5%	8%	10%	9%	8%
10	Atlanta	12%	14%	13%	17%	12%	11%	8%	7%	7%	8%	5%	3%
11	Miami	2%	3%	0%	1%	-1%	-4%	-6%	-8%	-11%	-13%	-11%	-10%
12	Minneapolis	-6%	-6%	-8%	-5%	-6%	-5%	-5%	-7%	-5%	-2%	-1%	-2%
13	Denver	5%	4%	1%	1%	2%	3%	1%	-1%	0%	1%	2%	3%
14	Salt Lake City	7%	7%	7%	13%	12%	8%	5%	9%	10%	9%	7%	6%
15	Raleigh	4%	2%	0%	5%	4%	3%	5%	5%	9%	11%	8%	6%
16	Boston	-5%	-5%	-3%	4%	-5%	-4%	-3%	-1%	1%	2%	3%	5%
17	Pittsburgh	-6%	-6%	-4%	-5%	-3%	-3%	-1%	-2%	-2%	-1%	-2%	-1%

Open the *SameStoreSales* data file and complete the following steps:

- Step 1.
 - Select cells B2:M17
- Step 2.
 - Click the Home tab on the Ribbon.
- Step 3.

- Click Conditional Formatting in the Styles group
- Select Colour Scales and click Blue-White-Red Colour Scale

Your heatmap should look similar to the below figure.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
2	St. Louis	-2%	-1%	-1%	0%	2%	4%	3%	5%	6%	7%	8%	8%
3	Phoenix	5%	4%	4%	2%	2%	-2%	-3%	-8%	-6%	-5%	-7%	-8%
4	Albany	-5%	-6%	-4%	-5%	-2%	-5%	-5%	-3%	-1%	-2%	-1%	-2%
5	Austin	16%	15%	15%	16%	18%	17%	14%	15%	16%	19%	18%	16%
6	Cincinnati	-9%	-6%	-7%	-3%	3%	6%	8%	11%	10%	11%	13%	11%
7	San Francisco	2%	4%	5%	8%	4%	2%	4%	3%	1%	-1%	1%	2%
8	Seattle	7%	7%	8%	7%	5%	4%	2%	0%	-2%	-4%	-6%	-5%
9	Chicago	5%	3%	2%	6%	8%	7%	8%	5%	8%	10%	9%	8%
10	Atlanta	12%	14%	13%	17%	12%	11%	8%	7%	7%	8%	5%	3%
11	Miami	2%	3%	0%	1%	-1%	-4%	-6%	-8%	-11%	-13%	-11%	-10%
12	Minneapolis	-6%	-6%	-8%	-5%	-6%	-5%	-5%	-7%	-5%	-2%	-1%	-2%
13	Denver	5%	4%	1%	1%	2%	3%	1%	-1%	0%	1%	2%	3%
14	Salt Lake City	7%	7%	7%	13%	12%	8%	5%	9%	10%	9%	7%	6%
15	Raleigh	4%	2%	0%	5%	4%	3%	5%	5%	9%	11%	8%	6%
16	Boston	-5%	-5%	-3%	4%	-5%	-4%	-3%	-1%	1%	2%	3%	5%
17	Pittsburgh	-6%	-6%	-4%	-5%	-3%	-3%	-1%	-2%	-2%	-1%	-2%	-1%

The cells shaded red indicate declining same-store sales for the month, and cells shaded blue indicate increasing same-store sales for the month.

Discuss how easy (or difficult) is it to interpret this heat map.

Activity 5 – Creating a Treemap using Excel

This activity creates a treemap using data from the *Brandvalues* data file. The data gives the top ten brand values for 2019 (source: Forbes.com). Each observation consists of an industry, a brand within an industry, and the value of the brand. The data is shown below.

	A	B	C
1	Industry	Brand	Value (\$ billions)
2	Technology	Apple	205.5
3	Technology	Google	167.7
4	Technology	Microsoft	125.3
5	Technology	Amazon	97.0
6	Technology	Facebook	88.9
7	Beverages	Coca-Cola	59.2
8	Technology	Samsung	53.1
9	Leisure	Disney	52.2
10	Automotive	Toyota	44.6
11	Restaurants	McDonald's	43.8

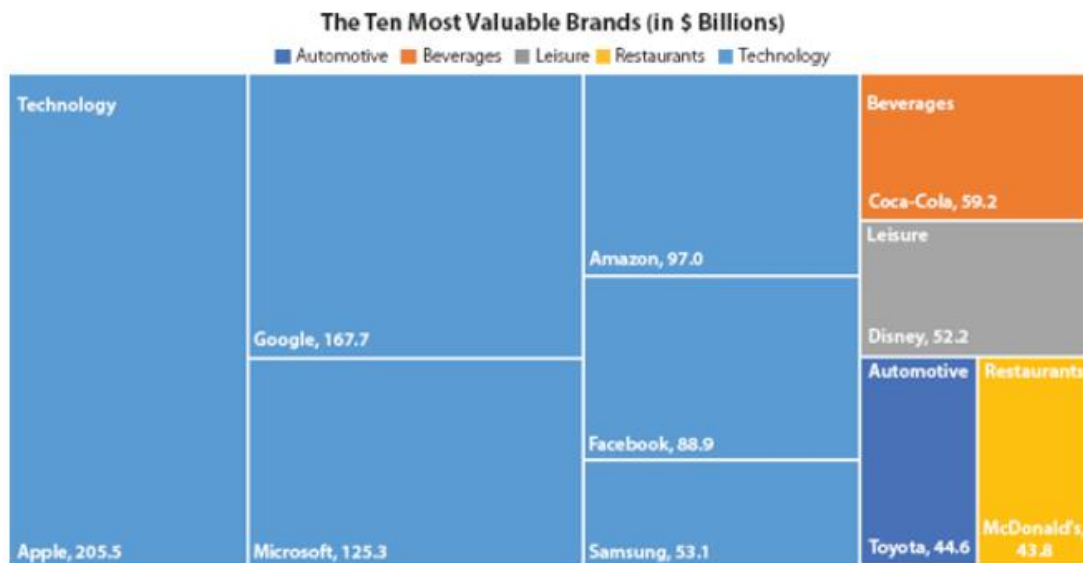
Open the *Brandvalues* data file and complete the following steps:

- Step 1.
 - Select cells A1:C11
- Step 2.
 - Sort the data by Industry by using the following steps:
 - Click Data on the Ribbon
 - Click the Sort button in the Sort & Filter group
 - In the Sort dialog box, select Industry from the drop-down menu
 - From the Order drop-down menu select A to Z
- Step 3.
 - Click Insert on the Ribbon.
 - Click the Insert Hierarchy Chart button in the Charts group
 - Select Treemap from the drop-down menu

To display the brand values:

- Step 4.
 - Click any brand label and then right click
- Step 5.
 - Select Format Data Labels... from the drop-down menu
- Step 6.
 - Select Text Options, Label Options and select Value

Your treemap should look similar to the below figure.



The colours in the treemap correspond to industries. Each of the rectangles represents a brand and the size of the rectangles indicates the size of the brand's value.

Activity 6 – Top Management

The Drucker Institute ranks corporations for managerial effectiveness based on a composite score derived from the following five factors: customer satisfaction, employee engagement and development, innovation, financial strength, and social responsibility. The file *ManagementTop25* contains the top 25 companies in the Institute's ranking based on the composite score (source: *The Wall Street Journal*). For each company, the industry sector, the company name, and the composite score are given.

1. Create a treemap chart using these data with the sector being the category, company being the subcategory, and the composite score being the quantitative variable. Use "Management Top 25" for the chart title. Hint: Be sure to first sort the data by sector.
2. In the sector with the most companies in this top 25, which company has the highest composite score?